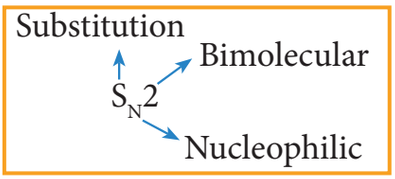
b) Unimolecular Nucleophilic substitution reaction (SN_1)

S_N2 Mechanism



The rate of S_N2 reaction depends upon the concentration of both alkyl halide and the nucleophile.

Rate of reaction

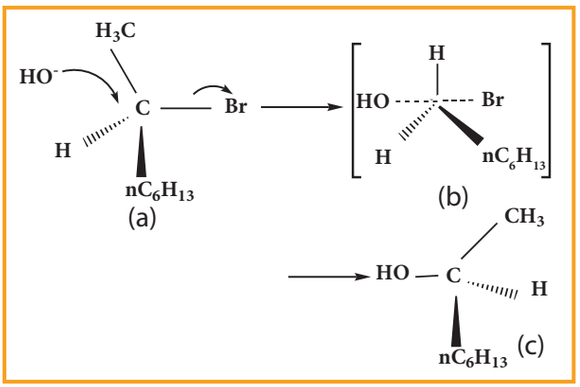
$= k_2$ [alkylhalide][nucleophile]

It follows second order kinetics and occurs in one step.

This reaction involves the formation of a transition state in which both the reactant molecules are partially bonded to each other. The attack of nucleophile occurs from the back side (i.e opposite to the side in which the halogen is attacked). The carbon at which substitution occurs has inverted configuration during the course of reaction just as an umbrella has tendency to invert in a wind storm. This inversion of configuration is called Walden inversion; after paul walden who first discovered the inversion of configuration of a compound in S_N2 reaction.

S_N2 reaction of an optically active haloalkane is always accompanied by inversion of configuration at the asymmetric centre. Let us consider the following example

When 2 - Bromooctane is heated with sodium hydroxide, 2 – octanol is formed with invesion of configuration. (-) – 2 – Bromo octane is heated with sodium hydroxide (+) – 2 – Octanol is formed in which – OH group occupies a position opposite to what bromine had occupied,



(a) (-) 2 – Bromo octane

(b) Transition State

(c) (+) 2 – Octanol (product)

S_N1 Mechanism

S_N1 stands for unimolecular nucleophilic substitution

'S' stands for substitution

'N' stands for nucleophilic

'1' stands for unimolecular (one molecule is involved in the rate determining step)

The rate of the following S_N1 reaction depends upon the concentration of alkyl halide (RX) and is independent of the concentration of the nucleophile (OH^-).

Hence Rate of the reaction

$= k$[alkyl halide]

$$R–Cl + OH^- \longrightarrow R – OH + Cl^-$$

This S_N1 reaction follows first order kinetics and occurs in two steps.

Unit 14 - jagan.indd 235 22-01-2021 13:04:58